

INDUSTRIAL INTERNSHIP REPORT

Prediction of Agriculture Crop Production in India

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Domain	Machine Learning
Internship Project	Prediction of Agriculture Crop Production in India
Program	upskill Campus / The IoT Academy / UCT

Final Project Report

EXECUTIVE SUMMARY

This report presents the work completed during the Machine Learning internship on the project "Prediction of Agriculture Crop Production in India." The project uses historical agricultural data to build a regression model for predicting crop production from previous-year Production, Area, and Yield information.

The yearly data was cleaned and converted into a structured machine-learning format. Previous-year values were created as input features and the following year's Production was used as the target. Linear Regression and Random Forest Regression were trained and compared.

Linear Regression was selected as the final model. On the test data it achieved an MAE of 28.32, RMSE of 40.86 and R^2 Score of 0.76. The completed project includes the notebook, saved model, processed data, prediction results, visualization and GitHub documentation.

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1 Preface

The internship provided an opportunity to work on a practical Machine Learning problem and understand the complete process of developing a data-driven solution. The project was completed through project understanding, dataset preparation, model development, testing, documentation and GitHub organization.

The selected problem, Prediction of Agriculture Crop Production in India, focuses on using historical agricultural information to estimate crop production. The project connected classroom concepts such as regression, train-test splitting, prediction and model evaluation with a practical dataset.

The internship also provided experience with Google Colab, Python, Pandas, NumPy, Scikit-learn, Matplotlib, Joblib and GitHub.

Internship Work Flow

- Understand the project and problem statement.
- Prepare and inspect the agricultural dataset.
- Clean and transform the data for machine learning.
- Create previous-year features and prepare the target.
- Train and compare regression models.
- Evaluate the models and select the final model.
- Test predictions and organize the project for submission.

2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

According to the supplied internship report template, UniConverge Technologies Pvt Ltd (UCT) was established in 2013 and works in the Digital Transformation domain, providing industrial solutions with a focus on sustainability and return on investment. The template describes UCT as using technologies including Internet of Things, Cyber Security, Cloud Computing, Machine Learning, communication technologies, Java Full Stack, Python and front-end technologies.

The template also describes UCT solutions including an IoT platform, smart factory solutions, LoRaWAN-based solutions and predictive maintenance applications.

2.2 About upskill Campus

The supplied template states that upskill Campus, along with The IoT Academy and in association with UniConverge Technologies, facilitated the internship process. It describes upskill Campus as a career development platform focused on personalized and scalable career development.

2.3 The IoT Academy

The supplied template describes The IoT Academy as the EdTech division of UCT and states that it runs executive certification programs in collaboration with EICT Academy and IIT institutions across multiple domains.

2.4 Objectives of the Internship Program

- Gain practical experience of working on an industry-oriented problem.
- Work on a real-world problem and develop a solution.
- Improve understanding of Machine Learning and its applications.
- Develop problem-solving and practical implementation skills.
- Improve career readiness and understanding of industry expectations.

2.5 References

- UCT internship project description and project requirements.
- Agriculture crop production dataset used for the project.
- Python, Pandas, NumPy, Scikit-learn, Matplotlib and Joblib documentation.

2.6 Glossary

Term	Meaning
ML	Machine Learning
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
R ²	Coefficient of Determination
CSV	Comma-Separated Values

3 Problem Statement

Agricultural production varies with crop type, cultivated area, yield and other factors. Historical production information can be used to identify patterns and support data-driven estimation of future production.

The assigned project focuses on predicting agriculture crop production in India using historical crop data. In this implementation, previous-year Production, Area and Yield values were used as input features, while the following year's Production was used as the prediction target.

Project Objectives

- Prepare historical agricultural data for machine learning.
- Create suitable previous-year input features.
- Train regression models to predict crop production.
- Compare model performance using standard regression metrics.
- Select and save a suitable final model.

Dataset Fields Used

Field	Role
Crop	Crop identifier
Year	Agricultural year
Production	Target production value
Area	Agricultural area
Yield	Crop yield
Previous Production	Input feature
Previous Area	Input feature
Previous Yield	Input feature

4 Existing and Proposed Solution

4.1 Existing Approach

The source agricultural data is organized by crop and year, with production, area and yield values provided for different years. In its original form, the yearly values are distributed across columns and are not directly convenient for the machine-learning workflow.

4.2 Proposed Solution

The proposed solution converts the yearly data into a structured format, creates previous-year features and trains regression models to predict the next year's production. Linear Regression and Random Forest Regression were tested.

4.3 Value Addition

- Transforms yearly data into an ML-friendly structure.
- Uses previous-year agricultural information as predictive features.
- Compares regression models before selecting the final model.
- Provides prediction results and an Actual vs Predicted visualization.
- Packages the implementation and supporting files in GitHub.

4.4 Code Submission (GitHub Link)

https://github.com/Srinath77-del/Agriculture-Crop-Production-Prediction/blob/main/agriculture_crop_prediction.ipynb

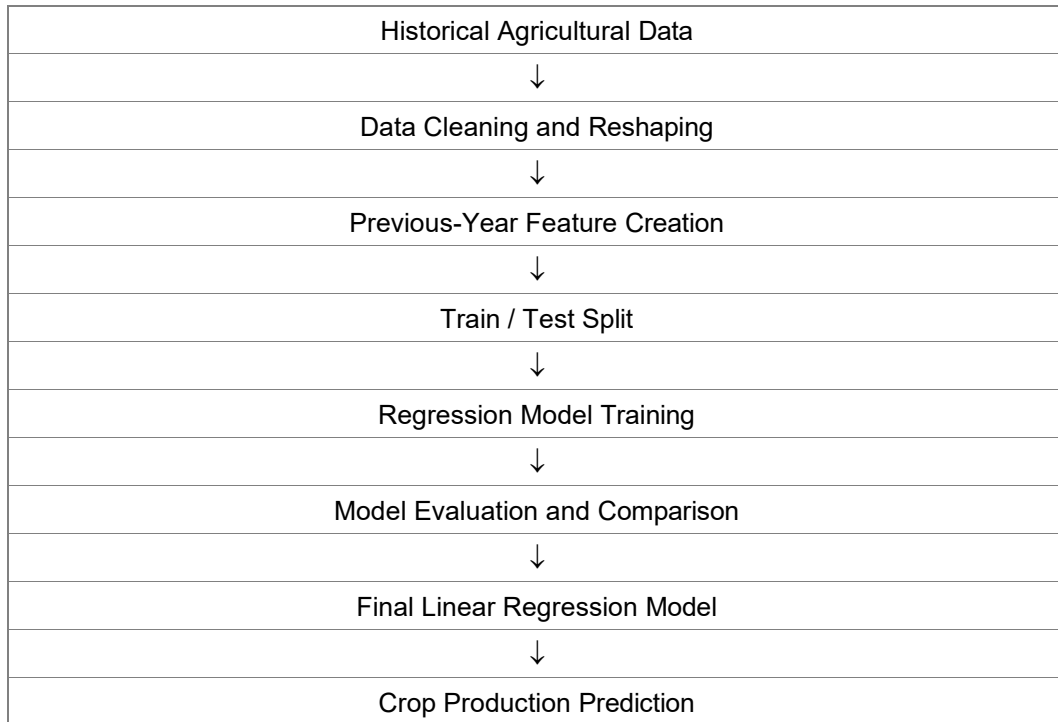
4.5 Report Submission (GitHub Link)

[Srinath77-del/Agriculture-Crop-Production-Prediction: Machine learning project for predicting agriculture crop production in India.](#)

5 Proposed Design / Model

The solution follows a simple machine-learning pipeline. Historical agricultural data is loaded and cleaned, yearly values are transformed into rows, previous-year values are created as model inputs, and the next year's Production becomes the target.

5.1 High Level Design



Model Inputs and Target

- Previous Production
- Previous Area
- Previous Yield
- Target: Production

The processed machine-learning dataset contained 220 records. The data was divided into 176 training records and 44 testing records.

6 Performance Test

Model performance was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and R^2 Score.

6.1 Test Plan / Test Cases

Test Case	Purpose	Outcome
TC01	Load and preprocess dataset	Completed
TC02	Create previous-year features	Completed
TC03	Train Linear Regression	Completed
TC04	Train Random Forest Regression	Completed
TC05	Compare model metrics	Completed
TC06	Generate production prediction	Completed
TC07	Load saved model and reuse it	Completed

6.2 Test Procedure

- Prepare the machine-learning dataset and separate inputs from the target.
- Split the data into training and testing sets.
- Train Linear Regression and Random Forest Regression models.
- Generate predictions on the test data.
- Calculate MAE, RMSE and R^2 Score.
- Compare results and select the final model.
- Test the saved model using realistic previous-year input values.

6.3 Performance Outcome

Model	MAE	RMSE	R ² Score
Linear Regression	28.32	40.86	0.76
Random Forest Regression	30.26	41.10	0.76

Linear Regression was selected because it achieved lower MAE and RMSE than Random Forest Regression while both models achieved an R² Score of 0.76.

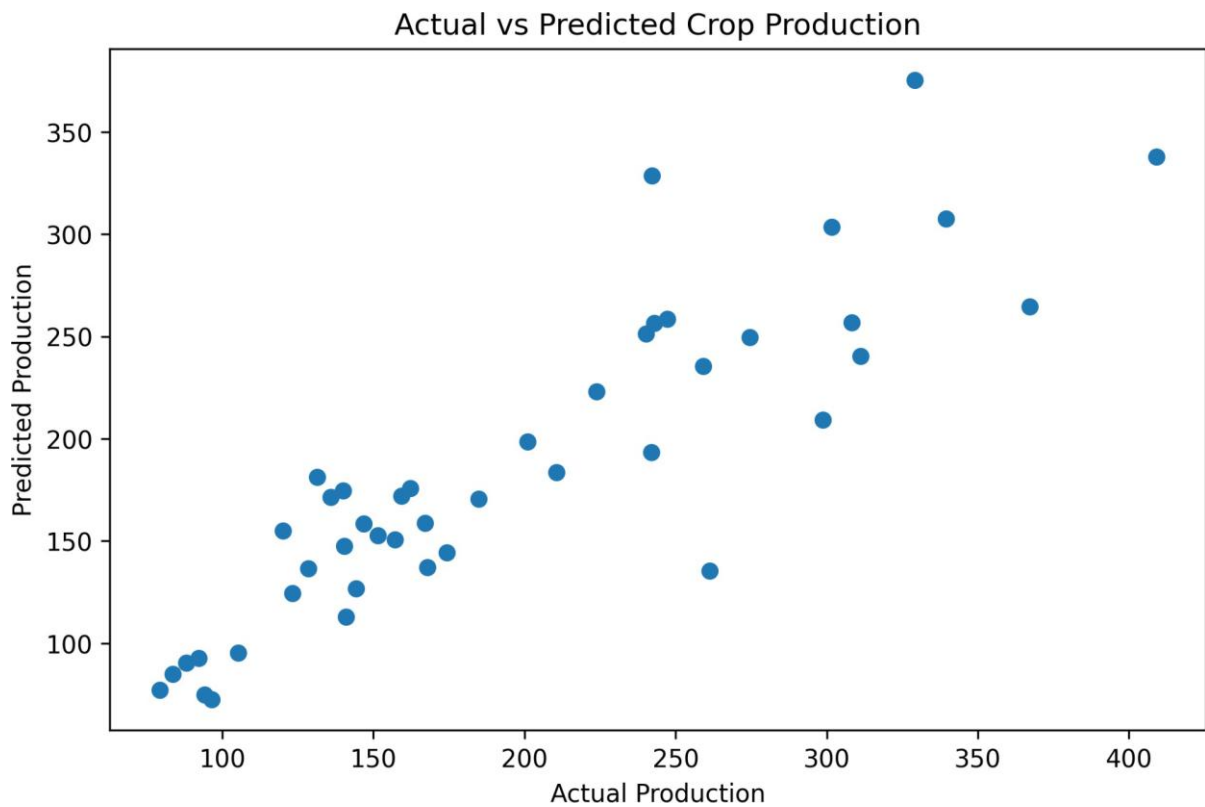


Figure 1: Actual vs Predicted Crop Production

7 My Learnings

This internship provided practical experience in building a machine-learning solution from a real-world style dataset. I learned that understanding and preparing the data is an important part of machine learning before model training begins.

- Inspecting and cleaning datasets using Pandas.
- Reshaping yearly data into an ML-friendly structure.
- Creating previous-year features for prediction.
- Splitting data into training and testing sets.
- Using Linear Regression and Random Forest for regression.
- Evaluating models using MAE, RMSE and R^2 Score.
- Saving and loading trained models using Joblib.
- Creating visualizations using Matplotlib.
- Organizing and documenting a project using GitHub.

These skills will help in future internships and projects because they provide experience with the complete workflow of a basic machine-learning project.

8 Future Work Scope

The current project demonstrates a basic production-prediction workflow using previous-year Production, Area and Yield. The solution can be extended by using additional agricultural and environmental information.

- Include rainfall and temperature information.
- Include soil and fertilizer-related information.
- Use more recent agricultural data and additional years.
- Include state or regional information when suitable data is available.
- Test additional regression algorithms.
- Perform hyperparameter tuning for tree-based models.
- Develop a simple web interface for prediction.
- Evaluate the system on a larger and more diverse dataset.

Conclusion

The project demonstrates how historical agricultural data can be processed and used to build a machine-learning regression model for crop production prediction. Linear Regression was selected as the final model with an R^2 Score of 0.76 on the test data. The completed project has been organized with its code, trained model, processed data, prediction results, visualization and documentation.